Hiloni Patel

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Education

Rutgers University New Brunswick, NJ
B.A. in Computer Science Expected January 2027
GPA: 3.61/4.00
Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Systems Programming, Computer Architecture, Software Methodology, Database Management, Internet Technology

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, C, SQL, OCaml, HTML/CSS
Frameworks and Technologies: React, React Router, Node.js, Express.js, Java Servlets, JDBC, Android, Apache Tomcat, REST APIs
Databases: PostgreSQL, MySQL, Relational Database Design
Developer Tools: Git, GitHub, npm, VS Code, IntelliJ IDEA, Android Studio, MySQL Workbench
Concepts: Object-Oriented Programming, Data Structures and Algorithms, Layered Architecture, Service/Repository Pattern, Request Validation, Error Handling, JWT Authentication, Role-Based Access Control, Automated Testing
Languages: English (Full Professional Proficiency), Gujarati (Native or Bilingual Proficiency), Hindi (Limited Proficiency)

Projects

TaskForge – Multi-Tenant Project Management Platform *React, TypeScript, Node.js, Express, PostgreSQL*

- Built a full-stack project management platform supporting organizations, memberships, projects, and task workflows through REST APIs.
- Designed 20 API route declarations across four Express modules covering authentication, organization membership, project management, and task operations.
- Implemented JWT authentication and organization-scoped role-based access control for OWNER, ADMIN, MEMBER, and VIEWER roles, enforcing permissions through backend middleware.
- Modeled five PostgreSQL tables with relational constraints and composite foreign keys to maintain organization-level data consistency across projects, tasks, and memberships.
- Developed task assignment, status, priority, due-date, sorting, and update workflows with backend validation and role-aware frontend controls.

RETRACE – Incident Replay and Debugging Platform *React, TypeScript, Node.js, Express, PostgreSQL, Jest*

- Built an incident debugging platform with 13 API route declarations and four relational tables for incidents, ordered HTTP events, replay runs, and request-level results.
- Engineered sequential HTTP replay to execute recorded requests, capture actual responses and execution durations, and report request-level PASS/FAIL results.
- Developed a response diff viewer comparing expected and actual HTTP status codes and JSON bodies to identify response mismatches and failed requests.
- Implemented replay history and run-to-run comparisons to monitor pass rates and identify regressions across executions.
- Wrote five passing Jest tests covering order creation, payment processing, intentional failures, invalid inputs, and unknown orders.

FlightSystem – Online Travel Reservation System *Java, Servlets, JDBC, MySQL, Apache Tomcat*

- Developed a flight reservation system using Java Servlets, JDBC, and MySQL to support flight search, ticket booking, cancellations, waitlists, and customer account management.
- Designed a 12-table relational schema covering users, flights, airlines, airports, aircraft, reservations, waitlists, and customer questions.
- Implemented role-specific workflows for customers, customer representatives, and administrators, including reservation management, flight administration, and account operations.
- Built flight search supporting one-way and round-trip itineraries with flexible travel dates and waitlists for fully booked flights.
- Developed administrative reports for monthly sales, revenue by flight, airline, and customer, and identification of top customers and frequently booked flights.